

TrES-2b

R-1704

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_150720 · created 2026-08-02T14:37:51+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457223.78588 +/- 0.00038 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.131 +/- 0.043
Transit depth (Rp/Rs)^2	1.73 +/- 1.12 [%]
Orbital Inclination (inc)	82.24 +/- 2.95
Airmass coefficient 1 (a1)	0.144 +/- 0.042
Airmass coefficient 2 (a2)	1.87 +/- 1.3
Scatter in the residuals of the lightcurve fit is	29.92 %
Best Comparison Star	#1 - [622, 188]
Optimal Aperture	3.0
Optimal Annulus	3.0
Transit Duration (day)	0.042 +/- 0.042

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.131 ± 0.043 vs 0.1254 (deviation 0.13σ)
Duration fit vs published	0.042 d vs 0.0746 d (deviation 0.77σ)
Mid-time O–C	-0.7 min
Depth SNR	3.0
Residual scatter	29.92 %

Light curve

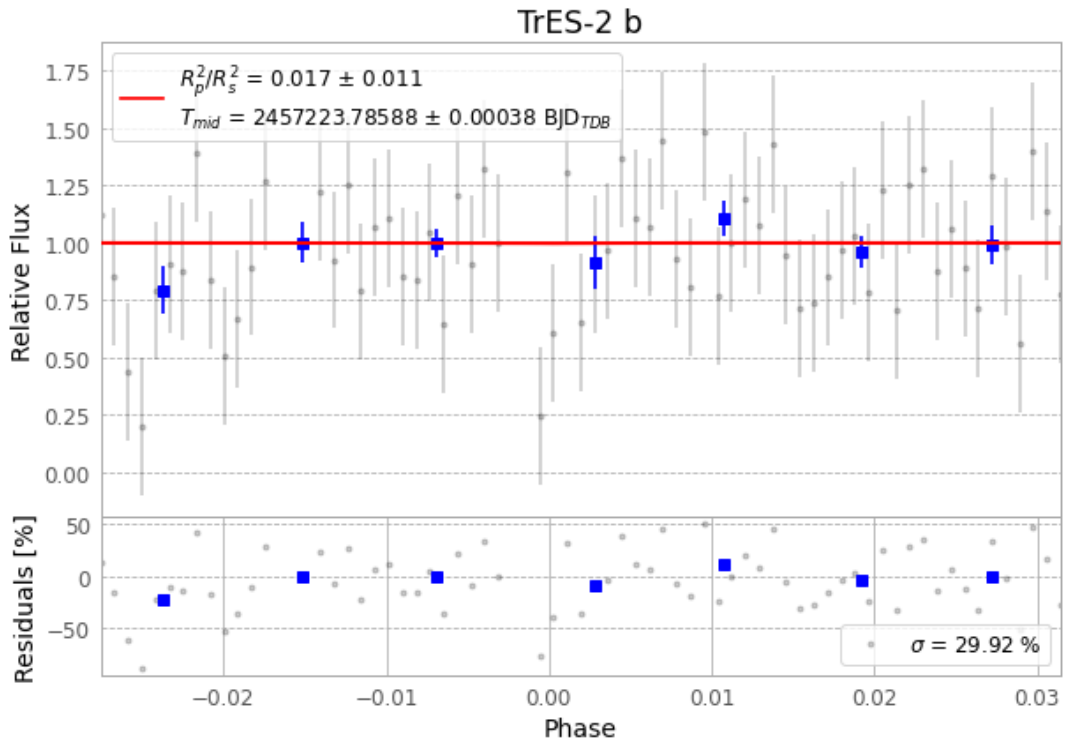


Figure 1 — FinalLightCurve_TrES-2 b_2015-07-19.png (SHA-256 9224eab2cff9948c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [594, 226], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7b41916d47b1bb0a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

71 FITS frames (47 MB), TRES-2150720050912.FITS → TRES-2150720083912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-150720.log (bef6751cb912...), FinalLightCurve_TrES-2 b_2015-07-19.png (9224eab2cff9...), FinalLightCurve_TrES-2 b_2015-07-19.csv (42605e660fac...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 14:37Z.